

Block 07

Cakici et al. (2024) — ML and the crypto cross-section

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MSCA Doctoral Network Digital Finance · MSCA_DF_10 Introduction to AI for Financial Applications



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Papers and materials

Cakici, N., Shahzad, S. J. H., Będowska-Sójka, B., & Zaremba, A. (2024).

Machine learning and the cross-section of cryptocurrency returns.

International Review of Financial Analysis, 94, 103244.

DOI [10.1016/j.irfa.2024.103244](https://doi.org/10.1016/j.irfa.2024.103244)

Notebook: notebooks/07_cakici_crypto_ml.ipynb

Siblings: 01a (Gu on equities) · 05 (crypto zombies: classifiers, not the return cross-section)

Paper card: Cakici, Shahzad, Będowska-Sójka & Zaremba (2024)

	<i>International Review of Financial Analysis 94: 103244</i>
Question	Can ML predict the cross-section of crypto returns, and does complexity help?
Data	500+ coins and tokens, 250 exchanges, weekly 2017–2023; 40 characteristics.
Method	Eight models: dimension reduction, OLS and regularized regressions, trees, neural nets; equal-weight forecast combination.
How it is tested	Predictive R^2 ; value-weighted long–short; three-factor alpha; costs; splits by limits to arbitrage.
Main result	Combination: 2.37% a week, Sharpe 1.66. Simple models win. Signal: price, past alpha, illiquidity, momentum.
Watch for	Profit is the long leg, in small / illiquid / volatile coins. Drawdowns of 70–80%.
Code & data	Coin panel not in our package. Classroom sketch is synthetic.

Why this paper next to 01a and 05

	Gu (01a)	Cakici	Zombies (05)
Market	equities	crypto	crypto
Object	$E[r]$	$E[r]$	delisting
Tool	ML ladder	ML ladder	classifiers
Complexity	usually helps	usually does not	RF vs logit

Same market as 05 . Same *question* as Gu — different answer on complexity.

Economic question

Is the crypto “factor zoo” a job for **deep** models, or do a few **simple** characteristics already sort winners from losers?

And once they do: is the alpha on the **short** leg (equities) or the **long** leg?

Data (paper vs classroom)

	Paper	Classroom sketch
Cross-section	500+ coins	80 synthetic coins
Frequency	weekly, 2017–2023	260 weeks
Characteristics	40	4 signal + 4 noise
Models	eight families	OLS, Ridge, RF, MLP, Combo

The planted signal is **linear** in price, alpha, illiquidity and momentum, and **stronger** for hard-to-trade names.

Eight families → four classroom models

OLS → Ridge → RF → MLP → Combo (equal-weight average)

Paper punchline: **OLS often beats trees and nets.**

Forecast combination is their best **value-weighted** Sharpe.

If the classroom R^2 table has OLS / Ridge on top, that is the point — not a bug.

Headline results (paper)

- All models make money; **complexity does not** add much
- Variable importance: **price, past alpha, illiquidity, momentum**
- Combination, value-weighted long–short: **2.37%** a week, Sharpe **1.66**, three-factor alpha **2.42%**
- R^2 looks disappointing; the **ranking** still works
- Gains **persist** in the second half of the sample (unlike many equity anomalies)

Long leg, not short leg

Equities: anomalies often live in the names you want to **short**.

Crypto (this paper): the pattern is **L-shaped**.

The top predicted quintile earns; the rest is mediocre.

So short-sale constraints are **not** the main story — **limits to arbitrage on the long side** are.

Costs and implementability

- High turnover, but most strategies **survive** costs (more than half the LS can disappear)
- Alphas **3–4×** larger in the hardest-to-trade tercile
- Four caveats they insist on:
 - i. 70–80% drawdowns
 - ii. tiny, illiquid coins
 - iii. signals die if you rebalance every two weeks (~30% less)
 - iv. you need the extreme volatile names

Same Avramov / Krauss lesson: *who* you are long is part of the result.

Live materials

Where	What
<code>notebooks/07_cakici_crypto_ml.ipynb</code>	synthetic panel; ladder; long vs short; VW / costs / hard-to-trade
Coin panel	not in the package

Map to the rest of the day

Earlier module	Bridge
Gu / 01a	same pipeline; opposite complexity result
Avramov et al.	ML alpha in hard-to-trade names
Krauss	a forecast is not a strategy without a portfolio rule and costs
Crypto zombies (05)	same market, different target

Do differently

1. Fit on the four signal columns only (drop the noise z^*)
2. Hold two weeks (paper: about 30% less profit)
3. Winsorize test returns at 1/99 (no extreme volatile coins)
4. Value-weight only the long leg

Tune on **validation**. Report on **test**.

Takeaway

A crypto ML cross-section is still the craft of the day:

simple characteristics → a model ladder → ranking, not R^2 → long vs short → who you can actually trade.

Gu asked whether complexity pays in equities. This paper's answer in crypto is: **usually not.**